HawkVision AI — Product Specification & Knowledge Base

Section 1: Executive Overview & Product Philosophy

HawkVision AI is an enterprise-grade artificial intelligence visual safety and operational monitoring system tailored for industrial, manufacturing, logistics, and heavy commercial environments. The platform transforms existing video camera networks (IP/RTSP) into real-time safety, security, and operational intelligence sensors using deep learning at the edge and centralized cloud analytics.

Supported Safety Usecases

- **Intrusion & Exclusion Zone Detection:** Identifies unauthorized personnel entering restricted areas, active heavy machinery paths, or dangerous operational zones. Supports time-scheduled rules and spatial sub-zoning.
- **PPE Compliance Monitoring:** Automatically verifies compliance with Personal Protective Equipment requirements, specifically detecting Hardhats, High-Visibility Vests, and Safety Glasses.
- **Vehicle Speeding & Logistics Control:** Measures real-time speed of forklifts, AGVs, semi-trucks, and yard vehicles across warehouse aisles, logistics yards, and pedestrian walkways.
- **Near-Miss Identification:** Uses spatial trajectory modeling to detect near-collisions between machinery/vehicles and ground personnel, enabling preventive intervention before actual incidents occur.
- **Fire & Smoke Detection:** Applies specialized vision models to identify early-stage flame ignition and smoke plumes in indoor facilities, chemical storage areas, and open yards faster than traditional thermal/particle sensors.

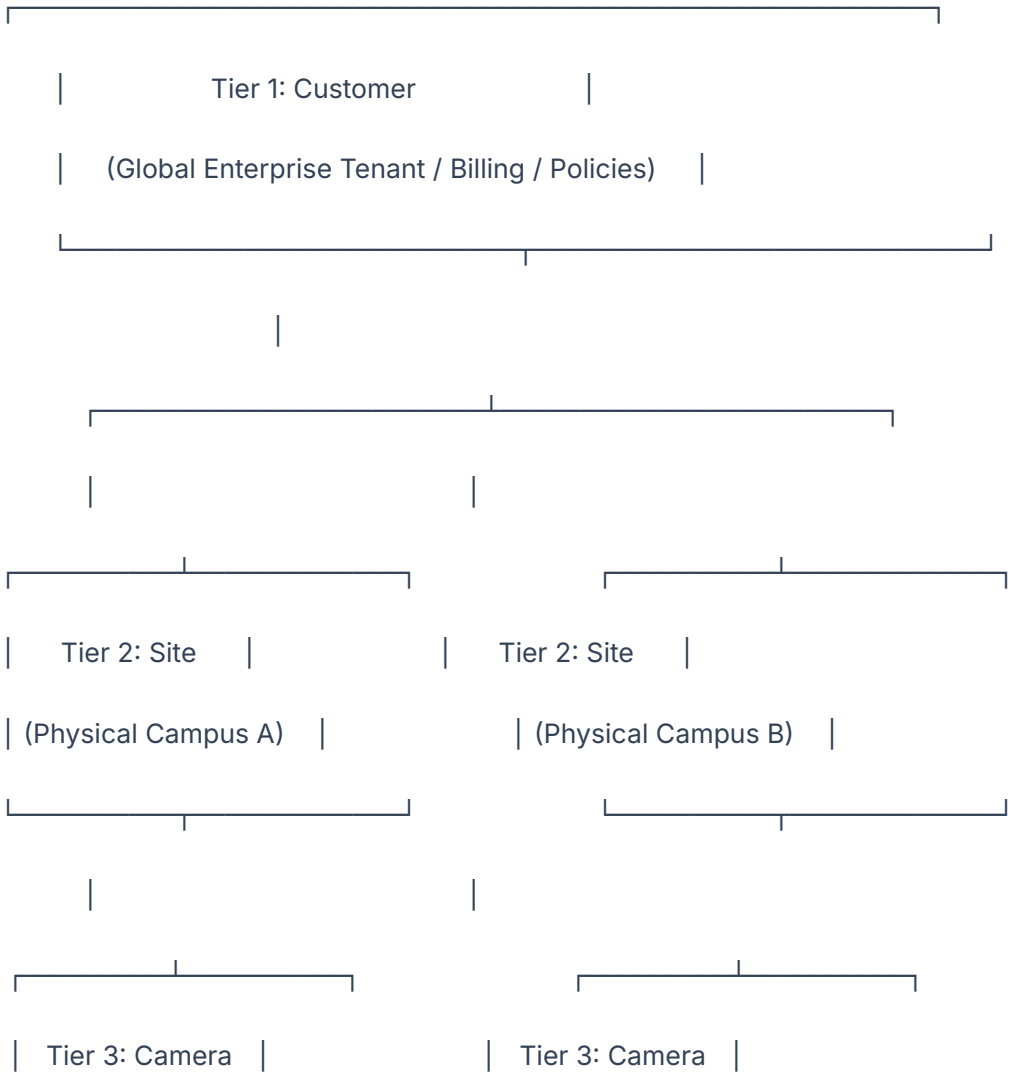
Core Design Principles

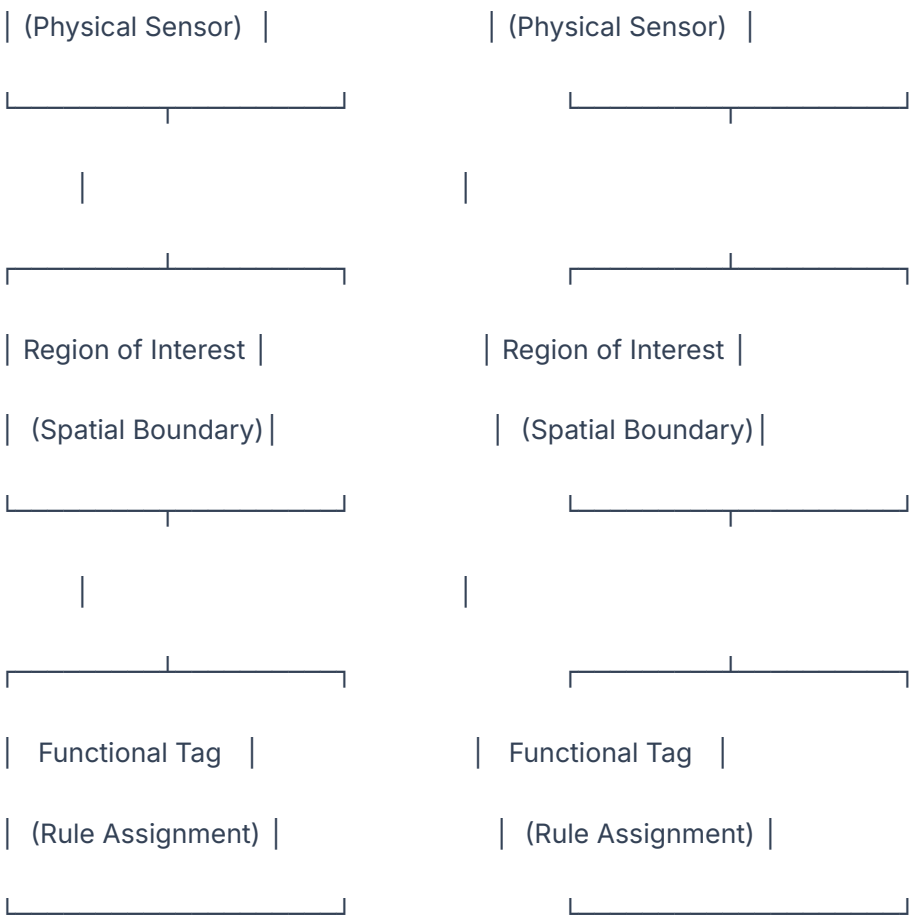
- **Distributed Edge Computing:** All computer vision inference, frame decoding, and spatial analysis occur locally on dedicated hardware nodes deployed at each facility site. Raw video streams never leave the local network, saving bandwidth and enforcing privacy.
- **Ultra-Low Latency Alerting:** Edge-level detection processes frame data in under 100 milliseconds, routing critical alerts immediately to site safety coordinators via push notifications, webhooks, or local sirens.

- **Multi-Tenant Hierarchy:** Features strict multi-level tenancy (Customer $\rightarrow$ Site $\rightarrow$ Camera $\rightarrow$ ROI) supporting granular access controls, global multi-site management, and isolated compute boundaries.
- **Automated Alert Routing & Triage:** Incident payloads containing annotated frames, metadata, and severity indexes are automatically categorized, delivered to site shift personnel, and monitored through a lifecycle tracking system.
- **Audit Compliance & Historical Safety Analytics:** Stores structured incident metadata and visual proofs to support OSHA, ISO 45001, and corporate EHS audit requirements, driving continuous risk mitigation strategies.

Section 2: Multi-Tenant Architecture & Data Model

HawkVision AI employs a strict three-tier organizational hierarchy designed to mirror multi-site enterprise operations while guaranteeing tenant data isolation and flexible administration.





Tier 1: Customer (Organization Level)

The top-level entity representing the enterprise organization (e.g., *Global Logistics Corp*).

- Controls global organizational settings, enterprise billing schedules, and master retention policies.
- Manages global administrator permissions (`HawkVision Admin`, `Org Admin`).
- Serves as the top-level isolation boundary for data tenancy, analytics aggregation, and license enforcement.

Tier 2: Site (Physical Campus / Facility Level)

Represents a distinct physical building, campus, manufacturing plant, or fulfillment center (e.g., *Austin Assembly Plant #4*).

- **Shift Configuration:** Defines local operational shifts (e.g., `Shift A - Day`, `Shift B - Night`, `24/7 Operations`). Incident logging and notification routing map directly to active shift schedules.
- **Site Functional Tags:** Holds the site-wide repository of Functional Tags tailored to local operational zones.
- **Edge Machine Association:** Links directly to local site hardware (Edge Node Appliance). Each site maintains a persistent sync connection with its edge machine.

Tier 3: Camera (Edge Visual Sensor Level)

Represents an individual physical visual sensor connected to the site network via RTSP (Real-Time Streaming Protocol) or ONVIF protocols.

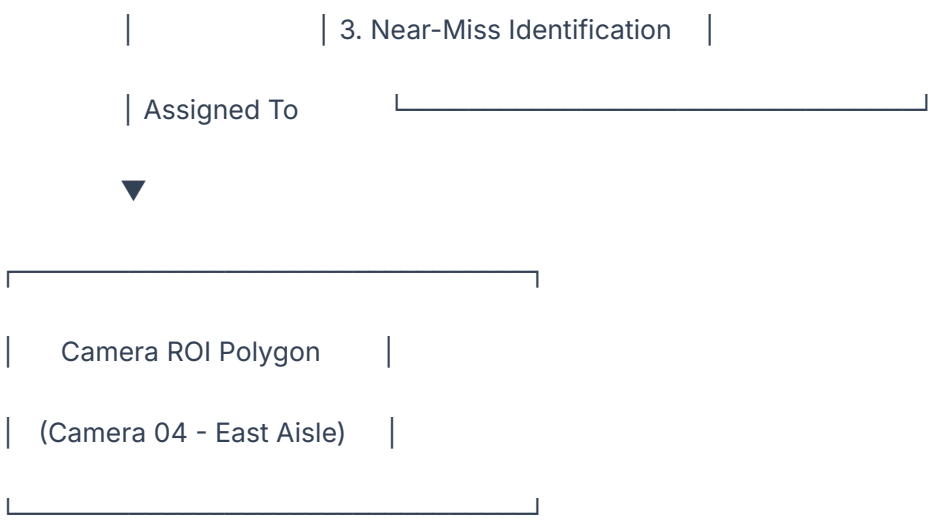
- Holds stream parameters: RTSP URL, resolution, target frame rate (FPS), network credentials, and stream health metrics.
- Contains one or more spatial Regions of Interest (ROIs).
- Houses the localized camera execution state (Online, Offline, Degraded).

Functional Tag Mechanism

Functional Tags serve as the core logical bridge between spatial camera regions and AI detection rules. Instead of manually binding computer vision models and threshold parameters directly to every individual camera frame, system administrators assign standardized operational classifications via Functional Tags.

- **Definition:** A Functional Tag represents a standardized zone type within an enterprise (e.g., `Forklift Aisle`, `Pedestrian Walkway`, `Chemical Storage Area`, `Loading Dock`).
- **Rule Mapping:** Specific AI Usecases (e.g., `Vehicle Speeding Detection`, `PPE Verification`, `Fire & Smoke`) are attached to a Functional Tag alongside operational thresholds (e.g., `speed_limit: 15km/h`, `required_ppe: ["hardhat", "vest"]`).
- **Execution Linkage:** When an operator draws an ROI polygon on a camera feed and assigns a Functional Tag to it, the ROI automatically inherits all AI detection rules tied to that tag.





Section 3: Interface & Feature Workflows

3.1 Home Tab (Dashboard & Analytics)

The Home Tab provides high-level visualization, operational telemetry, and incident aggregation across organizational levels.

Custom Dashboards & Layout Customization

Users can build role-based layouts by adding, resizing, and arranging modular widgets:

- **Incident Totals Metric Cards:** Real-time counts of total, critical, and unreviewed incidents across chosen sites.
- **Live Alert Ticker:** A real-time, streaming list of incoming edge detection events with instant preview thumbnails.
- **Severity Distribution Pie Charts:** Categorization of events into Low, Medium, High, and Critical (SIF - Serious Injury & Fatality) risk levels.
- **Site Comparison Bar Graphs:** Benchmark incident rates across different physical facilities to target safety training.
- **Time-Series Incident Trends:** Line charts showing safety trends over hours, days, weeks, or shifts.

Global Filter Bar

The Home Tab features a persistent multi-parameter filter bar that dynamically updates all rendered dashboard widgets:

- **Site:** Select all sites, a single campus, or custom site groups.

- **Camera:** Filter down to specific visual sensors.
- **Date & Time Range:** Absolute (custom calendar picker) or relative (`Last 1 Hour`, `Shift to Date`, `Last 30 Days`).
- **Shift:** Filter by defined operational schedules (`Shift A`, `Shift B`, `Night Shift`).
- **Severity Level:** Multi-select options (`Low`, `Medium`, `High`, `Critical`).
- **Usecase Type (`uctype`):** Filter by AI module (`PPE`, `Speeding`, `Fire`, `Intrusion`, `Near-Miss`).
- **Functional Tags:** Filter metrics by operational area types (e.g., all `Loading Dock` zones across the enterprise).

3.2 Configuration Tab (Sites, Cameras, ROIs, & Functional Tags)

The Configuration Tab serves as the central management engine for facility setup, spatial camera mapping, and rule definitions.

Site Management List

- Displays all configured sites within the customer organization.
- Shows connection status of edge nodes, connected camera counts, active shift profiles, and assigned location tags.

Functional Tags Sub-Tab

- **Creation & Editing:** Administrators define global or site-specific tags (e.g., `Hazardous Material Zone`).
- **Default Usecase Assignment:** Select default AI detection rules and parameter values (e.g., confidence thresholds, speed limits) tied to each tag.

Cameras Sub-Tab

- Lists all physical cameras linked to the selected site.
- **Status Monitoring:** Tracks sensor state using three indicators:
    - `Online`: Stream active, decoding normally at target FPS.
    - `Offline`: Stream unreachable or RTSP connection dropped.
    - `Degraded`: High frame drop rate, low lighting, camera obstruction, or severe latency.

Camera ROI & Usecase Configuration Workflow

Setting up visual detection on a camera follows a structured four-step workflow: ┌──┐


Step 1: Draw Region of Interest (ROI)
Draw custom polygon/rectangle over the live camera feed.
Step 2: Assign Functional Tag
Select a tag (e.g., "Main Transport Aisle") for the ROI.
Step 3: Dynamic Rule Fetching
System binds default AI usecases linked to the tag.
Step 4: Fine-tune & Save Rules
Toggle usecases, adjust thresholds, and save setup.

1. **Draw Region of Interest (ROI):** Open the camera setup view. Use the polygon or rectangle drawing tool to mark specific spatial boundaries on the live camera canvas.
2. **Assign Functional Tag:** Select a pre-configured Functional Tag from the dropdown list to apply to the newly created ROI.
3. **Dynamic Rule Fetching:** The UI automatically fetches and displays all default AI usecases and default detection thresholds tied to that tag.
4. **Fine-tune & Save Rules:** The operator can customize sensitivity settings, adjust thresholds, enable or disable specific usecases for this camera ROI, or remove existing rule assignments.

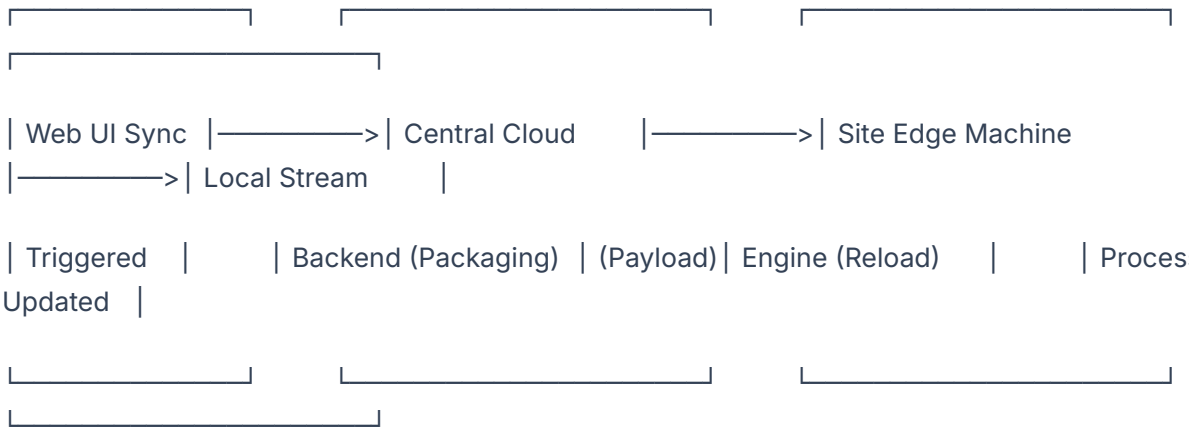
3.3 Site Edge Sync Mechanism & Technical Pipeline

Configuration changes made in the Cloud Web Application do not automatically disrupt live edge processing. They must be explicitly deployed using the Site Edge Sync mechanism.

Sync Button Functionality

Located in the header of every Site Configuration page. It turns amber whenever un-synced configuration changes exist (e.g., new ROIs drawn, camera added, tag modified). Clicking **Sync** initiates an atomic configuration push to the physical edge appliance deployed at that specific site.

End-to-End Sync Lifecycle



1. **Trigger:** User clicks **Sync to Edge** on the site configuration page.
2. **Packaging:** Central cloud backend compiles a unified JSON manifest containing site configurations, camera manifests, ROI coordinates, and active model parameters.
3. **Payload Delivery:** Cloud backend pushes the JSON payload over a secure WebSocket/TLS channel to the local Site Edge Machine.

4. **Edge Processing Reload:** Edge service parses the payload, updates local pipeline configs, updates local ROI spatial bounding masks, reloads active detection models, and resumes RTSP stream processing without dropping existing network streams.

Detailed Sync Payload Structure

The configuration package contains the following structure:

- **Site Identifiers & Metadata:** `site_id`, `site_name`, `customer_id`, timestamp.
- **Camera Manifest:** `camera_id`, `camera_name`, `stream_url`, `resolution`, `fps`.
- **ROI Mapping:** `roi_id`, coordinate point arrays `[[x1, y1], [x2, y2], ...]`, assigned `functional_tag_id`.
- **Attached Usecase Configurations:** `usecase_id`, `usecase_type`, `confidence_thresholds`, speed limit triggers, bounding parameters.
- **Rulebook & Shift Metadata:** Local operational shift schedules, notification escalation contacts, and severity scoring indices.

Edge-to-Backend Event Delivery Pipeline

1. **Video Decoding & Frame Capture:** Local edge machine ingests RTSP video streams from site cameras.
2. **Local AI Inference Engine:** Hardware acceleration passes frames through object detection, tracking, and spatial ROI mask algorithms.
3. **Violation Detection:** If an object (e.g., forklift without driver hardhat, speeding vehicle) crosses an ROI and breaks an active rule, an event is generated.
4. **Event Payload Construction:** Local edge machine builds an event package with bounding boxes, metric scores (e.g., speed), cropped snapshot images, and metadata.
5. **Event Transmission:** Edge machine securely transmits the light payload (JSON + cropped annotated image frame) back to the Cloud Backend Event Table.

3.4 Event Table & Incident Triage

The Event Table serves as the primary operational triage view for site safety managers, EHS officers, and security personnel monitoring safety incidents.

Real-Time Incident Table Structure

S.No / Event ID	Site Name	Camera Name	Usecase Type	Time Created	Received Time	Severity / Risk Level	Event Status	Event Details	Rulebook	Location Tags	User Shift
`EV-9021`	Plant Alpha -	Cam-04 East	Speeding	10:14:02 AM	10:14:03 AM	High	`Logged`	`speed: 34 km/h`	Logistics Rulebook	Loading Dock	Shift A -

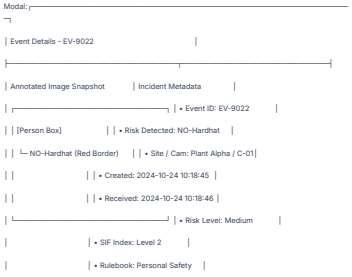
S.No / Event ID	Site Name	Camera Name	Usecase Type	Time Created	Received Time	Severity / Risk Level	Event Status	Event Details	Rulebook	Location Tags	User Shift
EV-9022	Plant Alpha -	Cam-01 West	PPE	10:18:45 AM	10:18:46 AM	Medium	Seen	NO-Hardhat	Personal Safety	Main Assembly	Shift A -
EV-9023	Yard South -	Cam-12 Gate	Intrusion	10:22:11 AM	10:22:12 AM	Critical	Invalid	Zone Intrusion	Perimeter	Security North	Night Shift -

Filtering & Sorting Options

- **Column Search:** Instant search on Event ID, Camera Name, and Event Details.
- **Alphabetical & Numeric Sorting:** Sort table columns A-Z, Z-A, or by ascending/descending timestamps.
- **Status Filtering:** Filter by `Logged`, `Seen`, or `Invalid`.
- **Date Range Picker:** Rapid filtering for daily, weekly, or custom shift audits.

Event Details Modal Workflow

Clicking on any row in the Event Table opens the Event Details Modal:





- **Annotated Image Snapshot:** Displays the frame captured at the exact moment of detection. Color-coded bounding boxes outline the offending object, person, or vehicle along with text labels (e.g., `NO-Vest`, `speed: 41 km/h`).
- **Metadata Review Panel:** Summarizes critical information: Risk Detected, Site Name, Camera Name, Created Time, Received Time, Risk Type, Detailed Parameters, SIF Index score, and active Rulebook.
- **Event Status State Machine:**
  - `Logged`: The default state automatically assigned when an incident payload is created by the edge and saved to the backend database.
  - `Seen`: The system automatically updates the event status from `Logged` to `Seen` as soon as an authorized user opens the Event Details Modal.
  - `Invalid`: If a safety officer determines the alert was a false positive, clicking the **Mark Invalid** button transitions the status to `Invalid`, excluding it from official safety violation metrics.

3.5 Users & Access Management Tab

Security and governance are controlled through granular role-based access control (RBAC).

User Roles & Access Rights Matrix

Role Name	Scope & Authority	Dashboard & Analytics	Site Config & Cameras	Functional Tags & ROIs	Edge Sync Trigger	User Management	Status Triage (Seen/Invalid)
HawkVision Admin	Global Platform Administrator (Cross-Tenant)	Full Access	Full Access	Full Access	Full Access	Full Access (Global)	Full Access
Org Admin	Customer Tenant Administrator	Full Access	Full Access	Full Access	Full Access	Full Access (Org Level)	Full Access
Site Admin	Single/Multi-Site Local Manager	Site Level	Site Level	Site Level	Site Level	Site Level (Local Users)	Full Access
Consumer	Field Operator / Safety Viewer	Read-Only	Read-Only	View Only	No Access	No Access	Update Status (Seen/Invalid)

User Table Attributes

The User Management interface displays:

- **Name & Email ID:** User identity credentials.
- **Role:** Assigned permission profile (`HawkVision Admin`, `Org Admin`, `Site Admin`, `Consumer`).
- **Job Title:** Organizational title (e.g., *Head of EHS*, *Plant Supervisor*).
- **Sites Assigned Count:** Number of physical facilities accessible to the user.
- **Location Tags:** Functional area restrictions (e.g., `Warehouse Alpha`, `Chemical Storage`).
- **Shift Schedule:** Assigned shift monitoring window (`24/7`, `M-F Day`, `M-F Night`).

Section 4: Data Schema & JSON Payload Specifications

4.1 Complete Sample JSON: Site Edge Sync Payload

This payload is generated by the Central Cloud Backend and transmitted to the Site Edge Machine when an operator triggers a **Sync**.{


```
 "sync_version": "2.4.0",

 "timestamp": "2024-10-24T08:30:00Z",

 "customer_info": {

  "customer_id": "cust-org-7721",

  "customer_name": "Apex Manufacturing Global"

 },

 "site_info": {

  "site_id": "site-austin-04",

  "site_name": "Austin Assembly Plant #4",

  "timezone": "America/Chicago",

  "edge_node_id": "edge-node-tx-01"

 },

 "camera_manifest": [
```

```
{
  "camera_id": "cam-8810-east",
  "camera_name": "East Logistics Corridor 01",
  "stream_url": "rtsp://admin:auth_token_secret@192.168.10.45:554/live/sub",
  "resolution": {
    "width": 1920,
    "height": 1080
  },
  "target_fps": 15,
  "status": "Online",
  "rois": [
    {
      "roi_id": "roi-9901-aisle",
      "roi_name": "Forklift Main Aisle Zone",
      "functional_tag_id": "tag-forklift-speed-zone",
      "functional_tag_name": "Forklift Heavy Traffic Aisle",
      "coordinates": [
        {"x": 120, "y": 240},
        {"x": 1800, "y": 240},
        {"x": 1850, "y": 950},
        {"x": 80, "y": 950}
      ],
```

```
"attached_usecases": [
 {
  "usecase_id": "uc-speed-01",
  "usecase_type": "Speeding",
  "enabled": true,
  "confidence_threshold": 0.85,
  "parameters": {
   "max_speed_limit_kmh": 12.0,
   "calibration_distance_meters": 25.5,
   "consecutive_frames_trigger": 3
  }
 },
 {
  "usecase_id": "uc-ppe-02",
  "usecase_type": "PPE",
  "enabled": true,
  "confidence_threshold": 0.80,
  "parameters": {
   "required_ppe_items": ["hardhat", "vest"],
   "ignore_seated_drivers": false
  }
```

```
        }
       ]
      }
     ]
    }
  ],
  "shift_metadata": [
    {
      "shift_id": "shift-day-a",
      "shift_name": "Shift A - Morning",
      "start_time_utc": "06:00",
      "end_time_utc": "14:00",
      "active_days": ["Mon", "Tue", "Wed", "Thu", "Fri"]
    }
  ]
}
```


4.2 Complete Sample JSON: Edge-to-Backend Event Generation Payload

This payload is generated by the Site Edge Machine and transmitted to the Cloud Backend Event Table when a safety violation is detected.


```
{
  "event_id": "EV-2024-88391",
  "schema_version": "1.2",
  "customer_id": "cust-org-7721",
```

```
"site_id": "site-austin-04",
"camera_id": "cam-8810-east",
"roi_id": "roi-9901-aisle",
"functional_tag_id": "tag-forklift-speed-zone",
"timestamps": {
  "created_time_edge": "2024-10-24T10:18:45.102Z",
  "received_time_cloud": "2024-10-24T10:18:45.310Z"
},
"usecase_type": "Speeding",
"severity": "High",
"sif_index": "Level-2-High-Risk",
"event_status": "Logged",
"rulebook": "Logistics & Yard Safety Code v3",
"event_details": {
  "primary_violation": "VEHICLE_SPEED_EXCEEDED",
  "detected_value": "24.5 km/h",
  "threshold_value": "12.0 km/h",
  "object_class": "forklift",
  "tracking_id": "trk-88219"
},
"bounding_box": {
  "x_min": 450,
```

```
    "y_min": 320,
    "x_max": 780,
    "y_max": 690,
    "confidence": 0.92
  },
  "media": {
    "annotated_snapshot_url":
"https://media.hawkvision.ai/snapshots/site-austin-04/2024/10/24/EV-2024-88391.jpg",
    "cropped_object_url":
"https://media.hawkvision.ai/crops/site-austin-04/2024/10/24/EV-2024-88391_crop.jpg"
  },
  "location_tags": ["Building A", "Logistics Bay 4"],
  "active_shift": "Shift A - Morning"
}
```


Section 5: RAG Evaluation Benchmark Questions

This evaluation suite contains 20 benchmark questions designed to test knowledge retrieval accuracy, structural hierarchy mapping, sync workflows, and permission enforcement for HawkVision AI systems.

1. Multi-Tenant Architecture & Structural Hierarchy

- **Question:** What are the three core structural tiers of the HawkVision AI organizational hierarchy, and what key configuration entities are defined at Tier 2?
- **Target Answer:** The three core structural tiers are Tier 1: Customer (Organization Level), Tier 2: Site (Physical Campus Level), and Tier 3: Camera (Edge Visual Sensor Level). At Tier 2 (Site), key configuration entities include site shift schedules, local site functional tags, and edge machine associations.

2. Functional Tag Mechanism

- **Question:** How do Functional Tags bridge spatial regions drawn on camera feeds with computer vision AI detection rules?
- **Target Answer:** Functional Tags act as logical middleware between spatial camera regions (ROIs) and AI usecase rules. Rather than binding detection parameters directly to a raw stream, rules and thresholds are bound to a Functional Tag. When an ROI polygon is drawn, assigning a Functional Tag causes the ROI to dynamically inherit all associated AI usecase parameters.

3. Edge Computing Philosophy & Privacy

- **Question:** Why does HawkVision AI perform computer vision frame decoding and AI model execution on local edge machines rather than transmitting live RTSP feeds directly to the cloud?
- **Target Answer:** HawkVision AI processes feeds on edge machines to achieve sub-100 millisecond detection latency for critical safety events, eliminate massive continuous cloud bandwidth requirements, and comply with privacy rules by keeping raw video streams on the local network. Only lightweight incident metadata and cropped snapshots are sent to the cloud.

4. Safety Usecases Supported

- **Question:** Name the five primary safety and operational usecases supported by HawkVision AI.
- **Target Answer:** The five primary usecases are: (1) Intrusion & Exclusion Zone Detection, (2) PPE Compliance Monitoring (Hardhat, Vest, Safety Glasses), (3) Vehicle Speeding & Logistics Control, (4) Near-Miss Identification, and (5) Fire & Smoke Detection.

5. Camera Status Definitions

- **Question:** What three status indicators represent camera health in the Configuration Tab, and what does the "Degraded" status specifically indicate?
- **Target Answer:** The three camera status indicators are `Online`, `Offline`, and `Degraded`. A status of `Degraded` indicates that the video stream is accessible but experiencing issues such as high frame drop rates, low lighting conditions, physical camera lens obstruction, or severe network latency.

6. ROI & Usecase Setup Workflow

- **Question:** Detail the 4-step workflow required to configure a new Region of Interest (ROI) and enable AI usecases on a camera feed.

- **Target Answer:** (1) Draw a rectangular or polygonal ROI boundary over the live camera frame, (2) Assign a Functional Tag to the newly created ROI, (3) System dynamically fetches default usecases attached to that tag, and (4) Operator fine-tunes thresholds, enables/disables specific usecases, and saves the setup.

7. Edge Sync Deployment Trigger

- **Question:** Why does an amber Sync button appear on the Site Configuration page, and what action occurs when a user clicks it?
- **Target Answer:** An amber Sync button indicates un-synced configuration changes (e.g., modified ROIs, newly added cameras, or updated functional tags). Clicking **Sync** compiles a complete JSON manifest and pushes it from the central cloud backend to the local site edge machine to reload its local detection engine without interrupting streams.

8. Detailed Site Edge Sync Payload Contents

- **Question:** Identify four primary data sections contained inside the Site Edge Sync payload JSON sent from the cloud backend to the edge machine.
- **Target Answer:** (1) Site Identifiers & Metadata (`site_id`, `customer_id`), (2) Camera Manifest (`stream_url`, resolution, target FPS), (3) ROI Mapping (coordinate polygons, `functional_tag_id`), and (4) Attached Usecase Configurations (usecase type, confidence thresholds, parameters).

9. Edge-to-Backend Event Generation Trigger

- **Question:** Describe the process that occurs on the local edge machine between frame ingestion and dispatching an incident to the backend Event Table.
- **Target Answer:** The local edge machine ingests raw RTSP frames, decodes them, and runs AI inference model passes. When an object crosses an ROI and breaks an active rule, the edge machine generates an event, constructs a JSON payload containing metadata and bounding boxes, captures an annotated frame snapshot, and posts it securely to the cloud backend.

10. Incident Lifecycle & Event Status Transitions

- **Question:** Explain the complete life cycle of an event status, detailing how an incident transitions from `Logged` to `Seen` and finally to `Invalid`.
- **Target Answer:** An event enters `Logged` status by default when created by the edge. When a user opens the Event Details Modal to review the incident, the system automatically transitions the status to `Seen`. If the user determines the alert was a false positive and clicks "Mark Invalid", the status manually changes to `Invalid`.

11. Event Details Modal Information

- **Question:** What visual elements and analytical metadata points are rendered inside the Event Details Modal?
- **Target Answer:** Visual element: The annotated image snapshot featuring bounding boxes and text labels around detected violations. Metadata points: Event ID, Risk Detected, Site Name, Camera Name, Created Time, Received Time, Risk Type, Specific Details (e.g., actual speed vs. limit), SIF Index score, and active Rulebook.

12. User Roles & Access Hierarchy

- **Question:** What are the four user roles supported in HawkVision AI, and which role possesses single-site local configuration capabilities without full organizational admin rights?
- **Target Answer:** The four user roles are `HawkVision Admin`, `Org Admin`, `Site Admin`, and `Consumer`. The `Site Admin` role manages single or multi-site local configurations without full organization-wide administrative privileges.

13. Consumer Role Permissions

- **Question:** What specific operations is a user with the `Consumer` role permitted to perform within the system?
- **Target Answer:** A `Consumer` has read-only access to assigned site dashboards and event feeds. They cannot modify site, camera, or ROI configurations, nor trigger Edge Syncs. However, they can interact with the Event Table to review incidents and update event statuses (`Seen`, `Invalid`).

14. Global Filter Bar Slicing Capabilities

- **Question:** List six parameters available in the Home Tab Global Filter Bar that allow operators to filter dashboard metrics.
- **Target Answer:** Parameters include: (1) Site, (2) Camera, (3) Date & Time Range, (4) Shift, (5) Severity Level (`Low`, `Medium`, `High`, `Critical`), (6) Usecase Type (`uctype`), and (7) Functional Tags.

15. Dashboard Analytics Widgets

- **Question:** Name four modular analytics widgets available for dashboard creation on the Home Tab.
- **Target Answer:** Widgets include: (1) Incident Totals Metric Cards, (2) Live Alert Ticker, (3) Severity Distribution Pie Charts, (4) Site Comparison Bar Graphs, and (5) Time-Series Incident Trends.

16. Shift Configuration & Operational Mapping

- **Question:** At which structural tier are shifts configured, and how do they influence incident logging?
- **Target Answer:** Shifts are configured at Tier 2 (Site Level). Incident payloads automatically capture the active shift schedule during event generation, allowing safety managers to aggregate and filter safety trends by specific work shifts (`Shift A`, `Shift B`, `Night`).

17. SIF Index Metric Purpose

- **Question:** What does the SIF Index measure in the Event Table and Event Modal metadata?
- **Target Answer:** SIF stands for Serious Injury & Fatality. The SIF Index scores the severity level of a safety violation based on its potential to cause life-altering harm, enabling safety teams to prioritize high-risk events over minor operational compliance slips.

18. User Table Attributes

- **Question:** Name five core user attributes tracked within the User Management interface table.
- **Target Answer:** Attributes include: (1) User Name, (2) Email ID, (3) Role, (4) Job Title, (5) Sites Assigned Count, (6) Location Tags, and (7) Shift Schedule.

19. Hardware Stream Parameters in Camera Manifest

- **Question:** Which camera stream parameters must be defined in the Camera Manifest within the Edge Sync Payload to ensure stream decoding?
- **Target Answer:** Parameters include: `camera_id`, `camera_name`, network `stream_url` (RTSP endpoint), target `resolution` (width and height), and target frame rate (`fps`).

20. Event Status Behavior on Metrics Reporting

- **Question:** How does marking an event as `Invalid` impact safety metrics reported across Home Tab widgets?
- **Target Answer:** Transitioning an event status to `Invalid` marks it as a false positive, removing it from official safety metric calculations, severity distribution pie charts, and compliance audit reporting.